Linear Algebra

MAT 226 — November 11th 2026

Homework 9

Q2) $w = \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$ $A = \begin{bmatrix} 2 & 6 & 4 \\ -3 & 2 & 5 \\ -5 & -4 & 1 \end{bmatrix}$

see if $A\vec{w} = \vec{0}$

$$\begin{bmatrix} 2 & 6 & 4 \\ -3 & 2 & 5 \\ -5 & -4 & 1 \end{bmatrix}\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = \begin{bmatrix} 2\times1 + 6\times-1 + 4\times1 \\ -3\times1 + 2\times-1 + 5\times1 \\ -5\times1 + -4\times-1 + 1\times1 \end{bmatrix}$$

$$= \begin{bmatrix} 2 - 6 + 4 \\ -3 - 2 + 5 \\ -5 + 4 + 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$A\vec{w} = \vec{0}$, hence, w is in Nul A.

Q6) $\begin{bmatrix} 1 & 3 & -4 & -3 & 1 \\ 0 & 1 & -3 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$

$R_1 \leftarrow R_1 - 3R_2$

$$\begin{bmatrix} 1 & 0 & 5 & -6 & 1 \\ 0 & 1 & -3 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$x_1 + 5x_3 - 6x_4 + x_5 = 0$$

$$x_2 - 3x_3 + x_4 = 0$$

$$x_1 = -5x_3 + 6x_4 - x_5$$

$$x_2 = 3x_3 - x_4$$

$$\vec{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} = \begin{bmatrix} -5x_3 + 6x_4 - x_5 \\ 3x_3 - x_4 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix}$$

$$= x_3 \begin{bmatrix} -5 \\ 3 \\ 1 \\ 0 \\ 0 \end{bmatrix} + x_4 \begin{bmatrix} 6 \\ -1 \\ 0 \\ 1 \\ 0 \end{bmatrix} + x_5 \begin{bmatrix} -1 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix}$$

The set of vectors that span Nul A are:

$$\left\{ \begin{bmatrix} -5 \\ 3 \\ 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 6 \\ -1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -1 \\ 0 \\ 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

8) $$\left\{ \begin{bmatrix} r \\ s \\ t \end{bmatrix} : 3r - 2 = 3s + t \right\}$$

$$3r - 2 = 3s + t$$
$$3r - 3s - t = -2$$

$$r = 0,\ s = 0,\ t = 0$$

$$0 - 0 - 0 \neq -2$$

There is no zero vector hence W is not a vector space.

16) $$\left\{ \begin{bmatrix} b-c \\ 2b+3d \\ b+3c-3d \\ c+d \end{bmatrix} : b, c, d \text{ real} \right\}$$

$$= \begin{bmatrix} b - c + d \\ 2b + 0c + 3d \\ b + 3c - 3d \\ 0b + c + d \end{bmatrix}$$

$$= b\begin{bmatrix} 1 \\ 2 \\ 1 \\ 0 \end{bmatrix} + c\begin{bmatrix} -1 \\ 0 \\ 3 \\ 1 \end{bmatrix} + d\begin{bmatrix} 1 \\ 3 \\ -3 \\ 1 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -1 & 1 \\ 2 & 0 & 3 \\ 1 & 3 & -3 \\ 0 & 1 & 1 \end{bmatrix}$$

26) (a) True

Null space is a valid vector space because it is a subspace of $\mathbb{R}^n$. Null space is the set of all solutions to $A\vec{x} = \vec{0}$

(b) True

An $m \times n$ matrix has m entries in each of its n columns. Because each column is in $\mathbb{R}^m$, their linear combination is also in $\mathbb{R}^m$.

(c) False

solutions to $A\vec{x} = \vec{b}$ is a solution set, not a column space

Col A is the set of all vectors $\vec{b}$ where $A\vec{x} = \vec{b}$ has a solution.

(d) True

This define Nul A

~~for~~ $T(x) = Ax$, where kernel is set of all x such that $A\vec{x} = \vec{0}$

(e) True

$T : A \to B$ is the set of all vectors in B

This range makes a column space for B

Column spaces are subspaces so the range is also a vector space.

(f) True

if $T(y) = 0$

then the solutions to the equation are those functions y that map to 0 when T is applied, which is exactly what a kernel is.

32) $T: P_2 \to \mathbb{R}^2$

$$T(p) = \begin{bmatrix} p(0) \\ p(0) \end{bmatrix}$$

let $p(t) = at^2 + bt + c$

for T check $t=0$

$$p(0) = a(0)^2 + b(0) + c = c$$

$$T(p) = \begin{bmatrix} c \\ c \end{bmatrix}$$

so c must be 0.

this leaves us

$$p(t) = at^2 + bt$$

polynomials that span kernel of T are

$$p(t) = t^2 \quad \& \quad p(t) = t$$

Range of T.

Applying T to any polynomial in P_2 gives us

$$\begin{bmatrix} c \\ c \end{bmatrix}$$

$$\begin{bmatrix} c \\ c \end{bmatrix} = c \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

Range of T is vector $\begin{bmatrix} 1 \\ 1 \end{bmatrix}$

34) T maps f to F where $F(0) = 0$

antiderivite means integral here

$$T(f)(x) = \int_0^x f(t)\,dt$$

linear Transformation

① addition $T(f+g)(x) = \int_0^x (f(t) + g(t))\,dt$

$$= \int_0^x f(t)\,dt + \int_0^x g(t)\,dt$$

$$= T f(x) + T g(x)$$

$$= T(f) + T(g)$$

② multiplication

$$T(cf)(x) = \int_0^x c f(t)\,dt$$

$$= c \int_0^x f(t)\,dt$$

$$= c\,T(f)$$

hence T is a linear transformation

kernel of T is the set of all functions that map to zero.

To find original input, take derivative of output

$$\int_0^x f(t)\,dt = 0 \quad \text{for} \quad x \in [0,1]$$

find f

$$\frac{d}{dx}\left(\int_0^x f(t)\,dt\right) = \frac{d}{dx}(0)$$

$$\boxed{f(t) = 0}$$

36) $T: V \to W$

show U is a subspace of V.

① $T(0_V) = 0_W$

0_W is in Z as Z is a valid subspace

$T(0_V)$ is in Z so 0_V is in U

② $T(x)$, $T(y)$

$T(x) + T(y)$ is in Z because of closure condition.
$T(x+y) = T(x) + T(y)$ (linear transformation)

since $T(x+y)$ is in Z, $x+y$ must be in U

③ take $T(x)$ & c

$cT(x)$ is in Z (closure condition)
$T(cx) = cT(x)$ (linear transformation)

$T(cx)$ is in Z so cx is in U

Hence U is a subspace of V.

38) $w = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 0 \end{bmatrix}$ $A = \begin{bmatrix} -8 & 5 & -2 & 0 \\ -5 & 2 & 1 & -2 \\ 10 & -8 & 6 & -3 \\ 3 & -2 & 1 & 0 \end{bmatrix}$

① check if w is in the Nul space of A

$$\begin{bmatrix} -8 & 5 & -2 & 0 \\ -5 & 2 & 1 & -2 \\ 10 & -8 & 6 & -3 \\ 3 & -2 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 1 \\ 0 \end{bmatrix}$$

$$= \begin{bmatrix} -8 \times 1 + 5 \times 2 + -2 \times 1 + 0 \times 0 \\ -5 \times 1 + 2 \times 2 + 1 \times 1 + -2 \times 0 \\ 10 \times 1 + -8 \times 2 + 6 \times 1 + -3 \times 0 \\ 3 \times 1 + -2 \times 2 + 1 \times 1 + 0 \times 0 \end{bmatrix}$$

$$= \begin{bmatrix} -8 + 10 - 2 + 0 \\ -5 + 4 + 1 - 0 \\ 10 - 16 + 6 + 0 \\ 3 - 4 + 1 + 0 \end{bmatrix}$$

$$= \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

w is in Nul A.

② check if $A\vec{x} = \vec{w}$ has atleast one valid solution

set up $[A|w]$

$$\begin{bmatrix} -8 & 5 & -2 & 0 & 1 \\ -5 & 2 & 1 & -2 & 2 \\ 10 & -8 & 6 & -3 & 1 \\ 3 & -2 & 1 & 0 & 0 \end{bmatrix}$$

$R_1 \leftrightarrow R_4$

$$\begin{bmatrix} 3 & -2 & 1 & 0 & 0 \\ -5 & 2 & 1 & -2 & 2 \\ 10 & -8 & 6 & -3 & 1 \\ -8 & 5 & -2 & 0 & 1 \end{bmatrix}$$

$R_2 \leftarrow R_2 + 2R_1$

$$\begin{bmatrix} 3 & -2 & 1 & 0 & 0 \\ 1 & -2 & 3 & -2 & 2 \\ 10 & -8 & 6 & -3 & 1 \\ -8 & 5 & -2 & 0 & 1 \end{bmatrix}$$

$R_2 \leftrightarrow R_1$

$$\begin{bmatrix} 1 & -2 & 3 & -2 & 2 \\ 3 & -2 & 1 & 0 & 0 \\ 10 & -8 & 6 & -3 & 1 \\ -8 & 5 & -2 & 0 & 1 \end{bmatrix}$$

$R_2 \leftarrow R_2 - 3R_1$, $R_3 \leftarrow R_3 - 10R_1$, $R_4 \leftarrow R_4 + 8R_1$

$$\begin{bmatrix} 1 & -2 & 3 & -2 & 2 \\ 0 & 4 & -8 & 6 & -6 \\ 0 & 12 & -24 & 17 & 19 \\ 0 & -11 & 22 & -16 & 17 \end{bmatrix}$$

$R_2 \leftarrow R_2/2$

$$\begin{bmatrix} 1 & -2 & 3 & -2 & 2 \\ 0 & 2 & -4 & 3 & -3 \\ 0 & 12 & -24 & 17 & 19 \\ 0 & -11 & 22 & -16 & 17 \end{bmatrix}$$

$R_3 \leftarrow R_3 - 6R_2$ $\qquad$ $R_4 \leftarrow R_4 + \frac{11}{2} R_2$

$$\begin{bmatrix} 1 & -2 & 3 & -2 & 2 \\ 0 & 2 & -4 & 3 & -3 \\ 0 & 0 & 0 & -1 & -1 \\ 0 & 0 & 0 & 1 & 1 \end{bmatrix}$$

$R_4 \leftarrow R_4 + R_3$, $R_3 \leftarrow -R_3$

$$\begin{bmatrix} 1 & -2 & 3 & -2 & 2 \\ 0 & 2 & -4 & 3 & -3 \\ 0 & 0 & 0 & +1 & +1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$R_1 \leftarrow R_1 + 2R_3$ $\qquad$ $R_2 \leftarrow R_2 - 3R_3$

$$\begin{bmatrix} 1 & -2 & 3 & 0 & 4 \\ 0 & 2 & -4 & 0 & -6 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & -2 & 3 & 0 & 4 \\ 0 & 1 & -2 & 0 & -3 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$R_1 \leftarrow R_1 + 2R_2$

$$\begin{bmatrix} 1 & 0 & -1 & 0 & -2 \\ 0 & 1 & -2 & 0 & -3 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

there is no row where $0\,0\,0\,0\,x$ where $x \neq 0$

so system is consistent.

$x_1 - x_3 = -2$ $\quad\quad$ $x_1 = x_3 - 2$

$x_2 - 2x_3 = -3$ $\quad\quad$ $x_2 = 2x_3 - 3$

$x_4 = 1$

x_3 is a free variable

let's take $x_3 = 1$ then

$x_1 = -1$
$x_2 = -1$
$x_3 = 1$
$x_4 = 1$

$$\begin{bmatrix} -1 \\ -1 \\ 1 \\ 1 \end{bmatrix}$$

The system is consistent with infinitely many solutions & I such valid solution is $\begin{bmatrix} -1 \\ -1 \\ 1 \\ 1 \end{bmatrix}$